

1. RIPE = R/I/P/E first-line

WHAT YOU SEE

Banner 'RIPE Tubers' on the wagon

WHAT IT ENCODES

First-line active TB regimen: Rifampin, Isoniazid, Pyrazinamide, Ethambutol — 2 months of all four, then 4 months of rifampin + isoniazid.

BOARD TRAP

Four drugs up front prevent resistance; never start fewer than four for active, drug-susceptible TB before DST.

2. Rifampin — RNA polymerase

WHAT YOU SEE

Gunslinger cluster (markers 7–9) at upper right

WHAT IT ENCODES

Rifampin inhibits bacterial DNA-dependent RNA polymerase (rpoB). Bactericidal; key sterilizing drug. rpoB mutation = rifampin resistance, a marker for MDR-TB on Xpert.

BOARD TRAP

Rifampin is a potent CYP450 INDUCER (lowers warfarin, OCP, PI levels) and turns secretions orange — both expected, not toxic-stop signals.

3. Rifampin orange secretions

WHAT YOU SEE

Orange-tinged figures (markers 10–11)

WHAT IT ENCODES

Harmless orange-red discoloration of urine, sweat, tears; permanently stains soft contacts. Rifampin can also cause thrombocytopenia and flu-like syndrome with intermittent dosing.

BOARD TRAP

Orange fluids are benign — reassure; do not stop the drug.

4. Isoniazid — mycolic acid synthesis

WHAT YOU SEE

Outlaw on the ground with sombrero (markers 32–34)

WHAT IT ENCODES

INH is activated by catalase-peroxidase (katG) and inhibits InhA, blocking mycolic acid synthesis in the cell wall. Bactericidal against replicating bacilli. katG mutation = INH resistance.

BOARD TRAP

INH is the CYP450 INHIBITOR (opposite of rifampin's induction) — raises phenytoin/warfarin levels.

5. Isoniazid → B6 / neuropathy

WHAT YOU SEE

'PYR' marker (33) left side near outlaw

WHAT IT ENCODES

INH depletes pyridoxine (vitamin B6) → peripheral neuropathy; co-administer B6 in pregnant, diabetic, alcoholic, malnourished, uremic, and HIV patients. Also causes hepatitis and drug-induced lupus.

BOARD TRAP

Treat/prevent the neuropathy with B6 — do not discontinue INH; INH overdose seizures are also reversed with pyridoxine.

6. Pyrazinamide — acidic milieu

WHAT YOU SEE

Horse with markers 15/26/35 lower left

WHAT IT ENCODES

PZA is converted by pncA to pyrazinoic acid and is active in the acidic intracellular/caseous environment, killing dormant bacilli — this is why it shortens therapy to 6 months.

BOARD TRAP

PZA works specifically in acidic conditions; pncA mutation = PZA resistance.

7. Pyrazinamide hepatotox + uric acid

WHAT YOU SEE

Central horse/figures (markers 26–29)

WHAT IT ENCODES

PZA is the most hepatotoxic first-line drug and raises uric acid (inhibits renal urate excretion) → arthralgias/hyperuricemia. Continue the drug for asymptomatic hyperuricemia.

BOARD TRAP

PZA hyperuricemia is usually not true gout and does not require stopping; stop only for hepatotoxicity.

8. Ethambutol — arabinosyl transferase

WHAT YOU SEE

Cowboy with horse, markers 28–31 right

WHAT IT ENCODES

Ethambutol inhibits arabinosyl transferase (embB), blocking arabinogalactan synthesis in the cell wall. Bacteriostatic; added mainly to cover possible resistance until DST returns.

BOARD TRAP

embB mutation = ethambutol resistance; EMB is the weakest of the four and is dropped once susceptibility is confirmed.

9. Ethambutol → optic neuritis

WHAT YOU SEE

Honeycomb/diamond fence pattern (visual cue, marker 30)

WHAT IT ENCODES

Dose-dependent optic neuritis with decreased visual acuity and red-green color blindness. Check baseline and monthly visual acuity/color vision; stop permanently if it develops.

BOARD TRAP

Visual changes = ethambutol — discontinue and do not rechallenge; usually reversible if caught early.

10. Hepatotoxicity trio (I, R, P)

WHAT YOU SEE

Central scene with stagecoach figures (markers 12–18)

WHAT IT ENCODES

INH, rifampin, and pyrazinamide are all hepatotoxic. Stop all if ALT/AST >3× ULN with symptoms or >5× ULN without; reintroduce one drug at a time, rifampin first.

BOARD TRAP

Ethambutol is the non-hepatotoxic drug — its toxicity is the eye, not the liver.

11. Streptomycin — aminoglycoside

WHAT YOU SEE

Biplane/train upper left (markers 16/26 'ZFT')

WHAT IT ENCODES

Streptomycin (alternative/second-line) is an aminoglycoside binding the 30S ribosome, inhibiting protein synthesis. Causes ototoxicity and nephrotoxicity.

BOARD TRAP

Streptomycin is absolutely CONTRAINDICATED in pregnancy (fetal ototoxicity) — unlike INH/RIF/EMB which are pregnancy-safe.

12. Bactericidal vs bacteriostatic

WHAT YOU SEE

Active gunfight scene center

WHAT IT ENCODES

Bactericidal first-line drugs: isoniazid, rifampin, pyrazinamide, streptomycin. Ethambutol is bacteriostatic.

BOARD TRAP

Knowing ethambutol is the lone bacteriostatic agent (and least active) is a frequent test discriminator.

13. Resistance genes (rpoB/katG/embB/pncA)

WHAT YOU SEE

Spread of outlaw figures across the scene

WHAT IT ENCODES

Resistance loci: rpoB (RIF), katG/inhA (INH), embB (EMB), pncA (PZA). Xpert detects rpoB to flag MDR-TB rapidly.

BOARD TRAP

MDR-TB = resistance to at least INH + RIF; isolated rifampin resistance is treated as MDR until proven otherwise.

14. Monitoring: LFTs, vision, uric acid

WHAT YOU SEE

Fence/ground baseline across bottom

WHAT IT ENCODES

Baseline and serial monitoring: LFTs (all hepatotoxic drugs), monthly visual acuity/color (EMB), and uric acid/symptoms (PZA). Counsel on orange fluids (RIF).

BOARD TRAP

Match the monitoring to the drug: eyes→EMB, liver→INH/RIF/PZA, neuropathy/B6→INH, ototoxicity→streptomycin.